

Step-by-Step Transition-Based Dependency Parsing

"Economic news had little effect on financial markets."

Introduction

- ▶ We analyze the sentence: "**Economic news had little effect on financial markets .**"
- ▶ We use a transition-based dependency parser.
- ▶ Three key actions:
 - ▶ **SHIFT**: Move a word from the buffer to the stack.
 - ▶ **LEFT-ARC**: Make the second-top element of the stack dependent on the top element.
 - ▶ **RIGHT-ARC**: Make the top element of the stack dependent on the second-top element.

Step 0: Initial configuration

Transition:

- ▶ **Stack:** [ROOT]
- ▶ **Buffer:** [Economic, news, had, little, effect, on, financial, markets, .]

Economic news had little effect on financial markets .

Step 1: SHIFT

Transition: Move "Economic" from buffer to stack.

- ▶ **Stack:** [ROOT, Economic]
- ▶ **Buffer:** [news, had, little, effect, on, financial, markets, .]

Economic news had little effect on financial markets .

Step 2: SHIFT

Transition: Move "news" from buffer to stack.

- ▶ **Stack:** [ROOT, Economic, news]
- ▶ **Buffer:** [had, little, effect, on, financial, markets, .]

Economic news had little effect on financial markets .

Step 3: LEFT-ARC (ATT)

Transition: "Economic" modifies "news" (ATT - Attribute).
Remove "Economic".

- ▶ **Stack:** [ROOT, news]
- ▶ **Buffer:** [had, little, effect, on, financial, markets, .]

The diagram shows the sentence "Economic news had little effect on financial markets ." with an arc labeled "ATT" connecting the word "Economic" to the word "news".
Economic news had little effect on financial markets .

Step 4: SHIFT

Transition: Move "had" from buffer to stack.

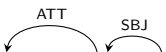
- ▶ **Stack:** [ROOT, news, had]
- ▶ **Buffer:** [little, effect, on, financial, markets, .]


Economic news had little effect on financial markets .

Step 5: LEFT-ARC (SBJ)

Transition: "news" is the subject of "had" (SBJ - Subject).
Remove "news".

- ▶ **Stack:** [ROOT, had]
- ▶ **Buffer:** [little, effect, on, financial, markets, .]

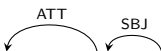


Economic news had little effect on financial markets .

Step 6: SHIFT

Transition: Move "little" from buffer to stack.

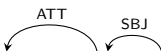
- ▶ **Stack:** [ROOT, had, little]
- ▶ **Buffer:** [effect, on, financial, markets, .]


Economic news had little effect on financial markets .

Step 7: SHIFT

Transition: Move "effect" from buffer to stack.

- ▶ **Stack:** [ROOT, had, little, effect]
- ▶ **Buffer:** [on, financial, markets, .]



Economic news had little effect on financial markets .

Step 8: LEFT-ARC (ATT)

Transition: "little" modifies "effect" (ATT - Attribute). Remove "little".

- ▶ **Stack:** [ROOT, had, effect]
- ▶ **Buffer:** [on, financial, markets, .]

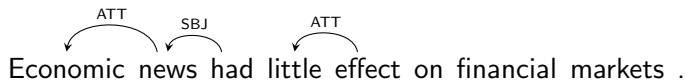
The diagram illustrates the LEFT-ARC (ATT) operation on the sentence "Economic news had little effect on financial markets .". It shows three arcs above the words: an arc from "little" to "effect" labeled "ATT", an arc from "had" to "effect" labeled "SBJ", and an arc from "effect" to "on" labeled "ATT".

Economic news had little effect on financial markets .

Step 9: SHIFT

Transition: Move "on" from buffer to stack.

- ▶ **Stack:** [ROOT, had, effect, on]
- ▶ **Buffer:** [financial, markets, .]



Step 10: SHIFT

Transition: Move "financial" from buffer to stack.

- ▶ **Stack:** [ROOT, had, effect, on, financial]
- ▶ **Buffer:** [markets, .]



Economic news had little effect on financial markets .

Step 11: SHIFT

Transition: Move "markets" from buffer to stack.

- ▶ **Stack:** [ROOT, had, effect, on, financial, markets]
- ▶ **Buffer:** [.]

Economic news had little effect on financial markets .

Step 12: LEFT-ARC (ATT)

Transition: "financial" modifies "markets" (ATT - Attribute).
Remove "financial".

- ▶ **Stack:** [ROOT, had, effect, on, markets]
- ▶ **Buffer:** [.]

The diagram illustrates the LEFT-ARC (ATT) transition on the sentence "Economic news had little effect on financial markets .". It shows four arcs connecting words in the sentence:

- An arc from "Economic" to "news" labeled "ATT".
- An arc from "had" to "news" labeled "SBJ".
- An arc from "little" to "effect" labeled "ATT".
- An arc from "on" to "markets" labeled "ATT".

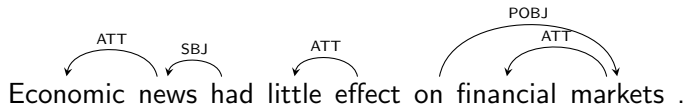
The sentence is written below the arcs: "Economic news had little effect on financial markets .".

Step 13: RIGHT-ARC (POBJ)

Transition: "markets" is the object of the preposition "on" (POBJ - Prepositional Object). Remove "markets".

► **Stack:** [ROOT, had, effect, on]

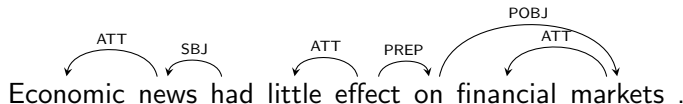
► **Buffer:** [.]



Step 14: RIGHT-ARC (PREP)

Transition: "on" functions as a preposition modifying 'effect' (PREP - Preposition). Remove "on".

- ▶ **Stack:** [ROOT, had, effect]
- ▶ **Buffer:** [.]

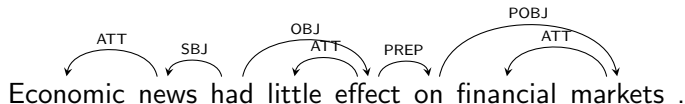


Step 15: RIGHT-ARC (OBJ)

Transition: "had" takes "effect" as its object (OBJ - Object).
Remove "effect".

► **Stack:** [ROOT, had]

► **Buffer:** [.]

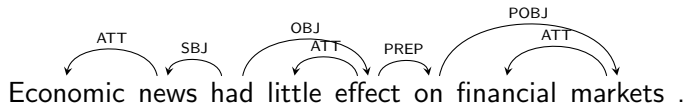


Step 16: SHIFT

Transition: Move "." from buffer to stack.

► **Stack:** [ROOT, had, .]

► **Buffer:** []

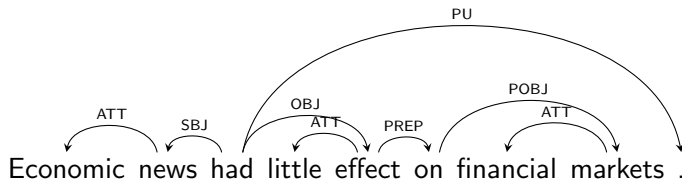


Step 17: RIGHT-ARC (PU)

Transition: "had" takes "." as punctuation (PU). Remove ".".

► **Stack:** [ROOT, had]

► **Buffer:** []



Step 18: RIGHT-ARC (PRED)

Transition: "ROOT" takes 'had' as predicate (PRED). Remove 'had'."

► **Stack:** [ROOT]

► **Buffer:** []

